

Yang Yan

Ph.D. Candidate, Zhejiang University / Westlake University

AI for Science | Computer Vision | Representation Learning | Structural Biology

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Research Profile

AI-for-science Ph.D. candidate developing foundation models and scalable ML systems for cryo-EM particle analysis and heterogeneous reconstruction. First author of a *Nature Methods* study introducing **Cryo-IEF**, the first foundation model for cryo-EM particle processing, pretrained on **65.3M particle images** and transferred to structural classification, pose-aware clustering, particle-quality ranking, and automated reconstruction. Current work develops CryoDECO, a foundation-prior framework for compositional and conformational heterogeneity in cryo-EM. Longer term, my research aims to build structural world models and agentic reconstruction systems that reason over molecular identity, dynamics, and native cellular context.

Education

2023–Present	Ph.D. Candidate in Computer Science Joint Ph.D. program supervised by Prof. Fajie Yuan and Prof. Huaizong Shen. Research: foundation models for cryo-EM image processing, automated workflows, and heterogeneity analysis.	Zhejiang University / Westlake University
2020–2023	M.S. in Translational Medicine (Engineering) National Institute of Diagnostics and Vaccine Development in Infectious Diseases, supervised by Prof. Ningshao Xia. Research: machine learning for medical image analysis.	Xiamen University
2016–2020	B.S. in Electrical Information Engineering Research included machine-learning applications in localization.	Northeastern University (Qinhuangdao)

Selected Publications

A comprehensive foundation model for cryo-EM image processing 2026

Yang Yan, Shiqi Fan, Fajie Yuan, Huaizong Shen

Nature Methods, 23(1), 88–95. DOI: [10.1038/s41592-025-02916-8](https://doi.org/10.1038/s41592-025-02916-8). Introduced Cryo-IEF as the first cryo-EM particle-processing foundation model and established a foundation-model paradigm for automated particle analysis.

Artificial intelligence foundation model automates cryo-EM structure determination 2026

Yang Yan, Huaizong Shen

Nature Methods Research Briefing, 23, 26–27, DOI: [10.1038/s41592-025-02917-7](https://doi.org/10.1038/s41592-025-02917-7).

CryoDECO: Deconstructing Extreme Compositional and Conformational Heterogeneity in Cryo-EM via Foundation Model Priors 2026

Yang Yan, Yanwanyu Xi, Shiqi Fan, Yifei Wang, Ziyun Tang, Fajie Yuan, Huaizong Shen

LangTaoSha Preprint Server. DOI: [10.65215/LTSpreprints.2025.12.30.000075](https://doi.org/10.65215/LTSpreprints.2025.12.30.000075). SOTA foundation-prior model for heterogeneous cryo-EM reconstruction, advancing high-throughput panoramic structural biology.

Unsupervised cryo-EM images denoising and clustering based on deep convolutional autoencoder and k-means++ 2022

D. Zhang, **Yang Yan**, Y. Huang, B. Liu, Q. Zheng, J. Zhang, N. Xia

IEEE Transactions on Medical Imaging, 42(5), 1509–1521. Deep autoencoder and balanced K-means++ framework for cryo-EM denoising and clustering.

Machine learning for structure determination in single-particle cryo-electron microscopy: A systematic review 2021

J. G. Wu, **Yang Yan**, D. X. Zhang, B. W. Liu, Q. B. Zheng, X. L. Xie, S. Q. Liu, S. X. Ge, et al.

IEEE Transactions on Neural Networks and Learning Systems, 33(2), 452–472. Review of ML methods for single-particle cryo-EM structure determination.

Selected earlier medical-imaging publications

2022–2023

S. Li, B. Liu, S. Li, X. Zhu, **Yang Yan**, D. Zhang; D. Zhang, B. Liu, Y. Huang, **Yang Yan**, et al.

Complex & Intelligent Systems, 2022; Journal of Digital Imaging, 2023. Deep-learning and ordinal-regression methods for bone-age assessment.

Research Highlights

- **First cryo-EM foundation model:** introduced **Cryo-IEF**, the first foundation model developed for cryo-EM particle processing, pretrained on **~65M particle images** and establishing a new paradigm for automated particle analysis in structural biology.
- **Automated reconstruction system:** developed **CryoWizard**, a fully automated computational pipeline for single-particle cryo-EM reconstruction that streamlines raw data processing, particle ranking, and high-resolution 3D structure determination without manual intervention.
- **Foundation-prior heterogeneous reconstruction:** developed **CryoDECO** to use pretrained particle representations as priors for ab initio heterogeneous reconstruction, targeting compositional and conformational mixtures.
- **Open-source Python package for cryo-EM data processing:** developed **cryodata**, a PyPI-installable data layer with **over 18.2k downloads** (according to pepy.tech) as of May 2026, converting CryoSPARC particle jobs into PyTorch-ready datasets with MRC/MRCS preprocessing, FFT/Hartley representations, LMDB storage, balanced sampling, and CryoSPARC-to-RELION metadata conversion.

Research Software and Open Source

2023–2026	Cryo-IEF: First Foundation Model for Cryo-EM Particle Processing https://github.com/westlake-repl/Cryo-IEF Designed and released the Cryo-IEF ecosystem, pretrained on ~65M cryo-EM particle images. Built PyTorch inference and downstream tooling for CryoRanker and CryoClustering, enabling practical structural classification, pose clustering, and particle-quality assessment.
2024–2026	CryoDECO: SOTA Heterogeneous Reconstruction with Foundation Model Priors https://github.com/yanyang1998/CryoDECO Developed a prior-guided heterogeneous reconstruction framework that uses Cryo-IEF representations to reduce random ab initio initialization and disentangle compositional classification from 3D reconstruction.
2024–2025	CryoWizard: Fully Automated Single-Particle Cryo-EM Reconstruction Pipeline https://github.com/SMART-StructBio-AI/CryoWizard Built and extended an end-to-end computational pipeline integrating CryoRanker with CryoSPARC, streamlining raw movies, micrographs, or particles to high-resolution 3D volumes without manual intervention through command-line, web, and Chrome-extension interfaces.
2023–2026	cryodata: Open-Source Python Package for Cryo-EM Data Processing https://github.com/SMART-StructBio-AI/cryoief-data Created the reusable data layer used by Cryo-IEF, CryoDECO, and CryoWizard to convert raw CryoSPARC particle outputs into PyTorch-ready training and inference datasets. The package supports MRC/MRCS preprocessing, LMDB-backed dataset creation, Fourier/Hartley feature generation, balanced sampling, PyTorch data loading, and CryoSPARC-to-RELION metadata conversion. Packaged and documented the toolkit for pip install cryodata , reaching over 18.2k downloads (according to pepy.tech) as of May 2026 and turning lab-specific preprocessing into reproducible open-source scientific ML infrastructure.

Technical Skills

ML / Vision	Self-supervised learning, representation learning, foundation models, clustering, autoencoders, generative/reconstruction models, image denoising.
Structural Biology	Single-particle cryo-EM, cryo-ET, CryoSPARC/RELION workflows, ChimeraX visualization, MRC/MRCS data formats.
Systems	Multiprocessing data pipelines, parallel GPU training/inference.

Research Agenda

I aim to build foundation-model priors and agentic reconstruction systems for cryo-EM/cryo-ET that infer molecular identity, conformational dynamics, and native cellular context from large-scale imaging data. My near-term focus is representation-guided heterogeneous reconstruction; my long-term goal is AI-guided panoramic structural biology for complex molecular mixtures.